Measure Big Five Inventory Convergence with Word Embedding Models and Generative AI

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Abstract

Traditional methods for measuring the validity of psychometric tests can be a complicated procedure. It requires a resource-intensive data collection process while involving repetitive statistical analysis. This study proposes alternative approaches using language models including word embedding models and generative AI to measure the convergence of the Big Five Inventory (BFI). For word embedding models like GloVe, we could calculate similarity between test items by vectorizing their representations. For generative AI models such as ChatGPT, we will ask the model to assess similarity values between all pairs of items. Based on the similarity matrix, clustering methods will be used to measure the convergence of the five personality categories. The proposed methods streamline the analysis of semantic relationships between elements and can also enable the generation of new elements, providing a scalable and objective framework for the development and evaluation of psychometric tests.

Introduction

Psychologists have long attempted to establish scientific ways to understand and measure psychological phenomena. To ensure that the measurements assess their intended psychological constructs, test validity becomes a crucial metric. However, the process of creating and validating such tests can be highly resource-intensive. It often requires substantial participation from diverse groups to ensure representative data, as well as the application of complex statistical techniques like factor analysis to validate test structure and consistency. In the field of personality psychology, the development of the revised NEO Personality Inventory (NEO PI-R), introduced by Costa and McCrae, exemplifies this process. It was initially conceptualized in 1978 and finalized in 2005 after extensive refinements and validation efforts (APA, 2018; Boyle, Saklofske, & Matthews, 2015).

Meanwhile, multiple studies have introduced the usage of language models in analyzing personality structures (Reise, 2003; Ren, 2021; Bimler, 2007). In the review by Cutler et al, researchers investigated if Natural Language Processing (NLP) models can replicate personality traits described by the Big Five personality theory and the lexical hypothesis (Cutler & Condon, 2022). They demonstrated promising results, where NLPs can approximate traditional survey-based methods and extract traits like Agreeableness, Extraversion, and Conscientiousness with high accuracy. This approach offers scalability and the ability to analyze thousands of descriptors across languages without needing a large pool of participant responses. At the same time, researchers have also investigated the potential to use language models in measuring psychometric test validities (Fang, 2022; Wang, 2019).

Research Questions

Past research has demonstrated the ability of language models to extract personality traits from large text corpora and highlighted the potential of embedding models for validating psychometric constructs. This study seeks to explore the intersection of these capabilities by deploying language models to analyze the convergence of personality traits within the Big Five Inventory (BFI). It aims to address the following research questions: Will word embedding models and generative AI confirm convergence of BFI dimensionality validity? Will GenAI be able to produce new test items that also converge to the expected category?

Methods

There are numerous personality assessments based on the Big Five theory. As a preliminary step, this study focuses on the BFI, which consists of 44 items. These items are evenly distributed across the five personality traits, with 8 items each for Extraversion, Agreeableness, and Openness, and 9 items each for Conscientiousness and Neuroticism. Additionally, the inventory includes reverse-coded items to minimize response biases and enhance the reliability of the assessment. The list of question items were encoded by their categories as shown in table 1.

Traits	Encodings
Extraversion	0
Agreeableness	1
Conscientiousness	2
Neuroticism	3
Openness	4

Embedding Model: GloVe

This study utilized the GloVe embedding model, specifically the glove.42B.300d version, to calculate the vector representation of each item in the Big Five Inventory. Each item was first

preprocessed and tokenized to remove neutral or function words, such as “I” and “is”. For example, the sentence “I see myself as someone who is talkative” was transformed into the key terms “see”, “someone”, and “talkative”. These preprocessed tokens were then vectorized using GloVe, and the mean of the token vectors was computed to generate a single vector representation for each inventory item. Finally, cosine similarity was calculated between all pairs of vectors to create a similarity matrix, representing the semantic relationships between items.

Generative AI: ChatGPT

This study also used ChatGPT, specifically the GPT-4o model, to assess the semantic similarity between the 44 items in the BFI. Each pair of items was evaluated using a prompt designed to generate a similarity score. The structure was structured as follows: “Rate the semantic similarity between these two questions on a scale of -1 (completely different) to 1 (identical): 1. {question 1} 2. {question 2} Provide only the similarity score as a number”. For every pair of items, the prompt was called 30 times using GPT-4o’s API to reduce the variability in the model’s outputs, and the mean of the 30 scores was calculated as the final score for each pair. This process resulted in a complete similarity matrix, which represents the semantic relationships between all items in the inventory.

To the third research question, ChatGPT was also utilized to generate 5 new test items given the previous exposure to the 44 items in the BFI. The similarity matrix was then updated to include the 5 new items following the same procedure.

Clustering and Evaluation

The study used both spectral clustering and hierarchical clustering methods to assess whether the vectorized Big Five Inventory (BFI) items naturally converge into their intended

personality categories. Spectral clustering was applied directly to the similarity matrix, while hierarchical clustering utilized a distance matrix derived by subtracting each similarity value from 1. To analyze the clustering results, the items were encoded into clusters labeled from 0 to 4. A contingency table was then created to map each cluster to the most frequently occurring personality category within it. For instance, if a cluster had a row in the contingency table represented as [0, 0, 3, 0, 1], it would be categorized as conscientiousness, corresponding to the encoded index of 2, as this category has the highest occurrence in that cluster. The accuracy of clustering convergence to the true categories was measured by the percentage of correctly assigned items in the 44 items.

Results

ChatGPT outperformed GloVe in both clustering methods. In spectral clustering, ChatGPT achieved an accuracy of 0.70, compared to 0.57 for GloVe. In hierarchical clustering, GloVe identified two major clusters, mapping them to Extraversion and Openness, with accuracy slightly better than random chance. In contrast, ChatGPT successfully identified all five categories with a significantly higher accuracy 0.66, demonstrating its superior ability to capture the semantic relationships between items and align them with the intended personality traits.

GloVe

The similarity matrix ranged from 0 to 1, with 0 indicating an orthogonal relationship between item pairs (visualized in blue) and 1 indicating high similarity (visualized in red). Although cosine similarity typically ranges from -1 to 1, the similarities derived from GloVe were all positive. This could be attributed to the averaging process used to create vector representations for each item, which likely smoothed the representations and reduced variability, resulting in more homogenous outputs. However, this didn’t negatively impact the measure of

reversed items. Words describing the same construct, even when reversed, maintained higher similarity scores compared to words from different categories. For instance, "organized" shared a higher similarity score with "disorganized" than with "talkative." When reversed items were excluded from the analysis, the overall accuracy remained unchanged at 0.57, indicating their minimal influence on the clustering performance.

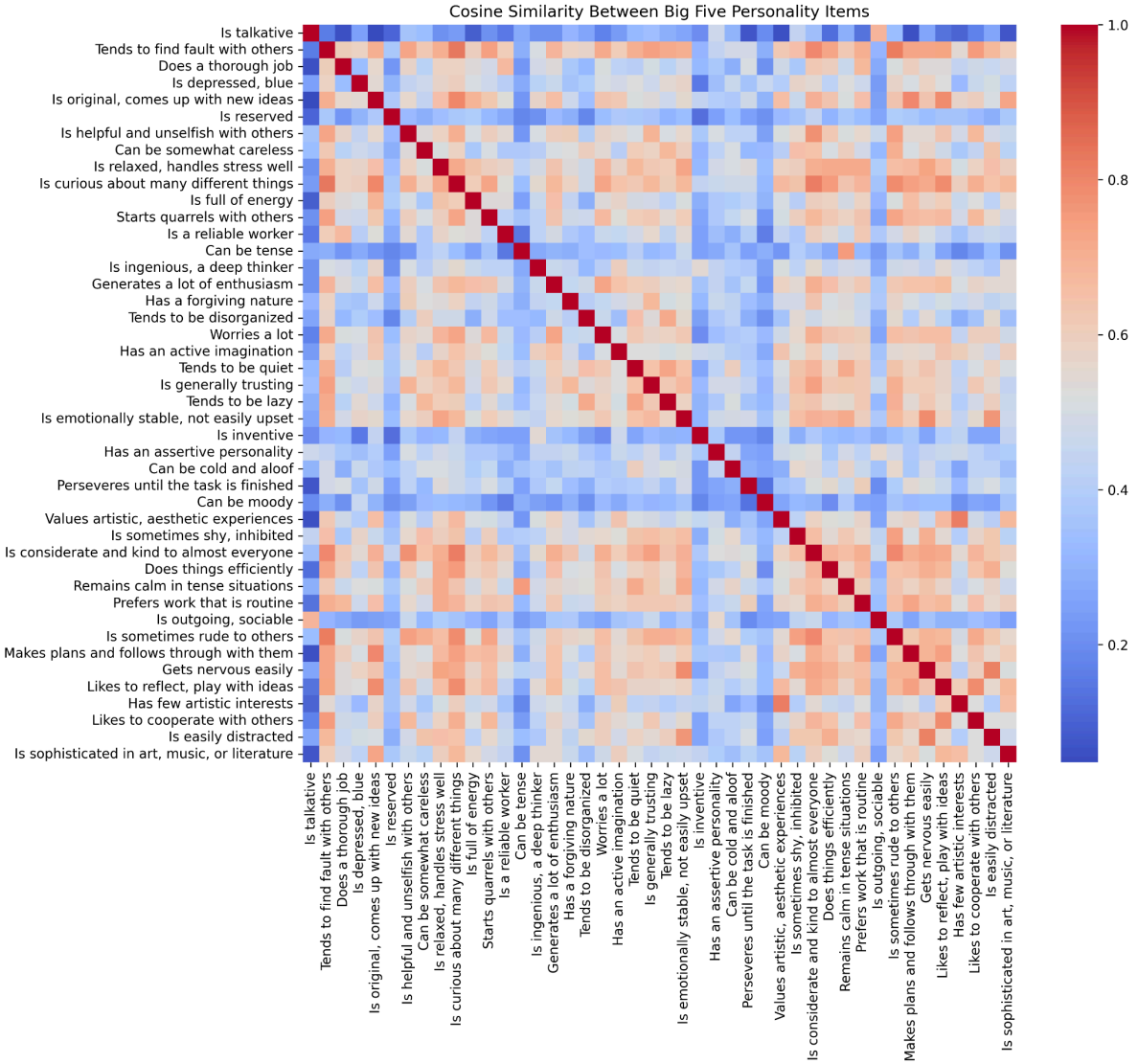


Figure 1: Similarity Matrix Generated by GloVe

In the hierarchical clustering analysis, GloVe resulted in two primary clusters: one for Extraversion (*N=7*) and another for Openness (*N=37*). This aligns with existing literature, which

is consistent with the literature where it was more obvious for language models to extract extraversion openness and agreeableness (Cutler & Condon, 2022). The hierarchical clustering result was transformed into five flat clusters to match the five personality traits. However, it gave the lowest accuracy of 0.43.

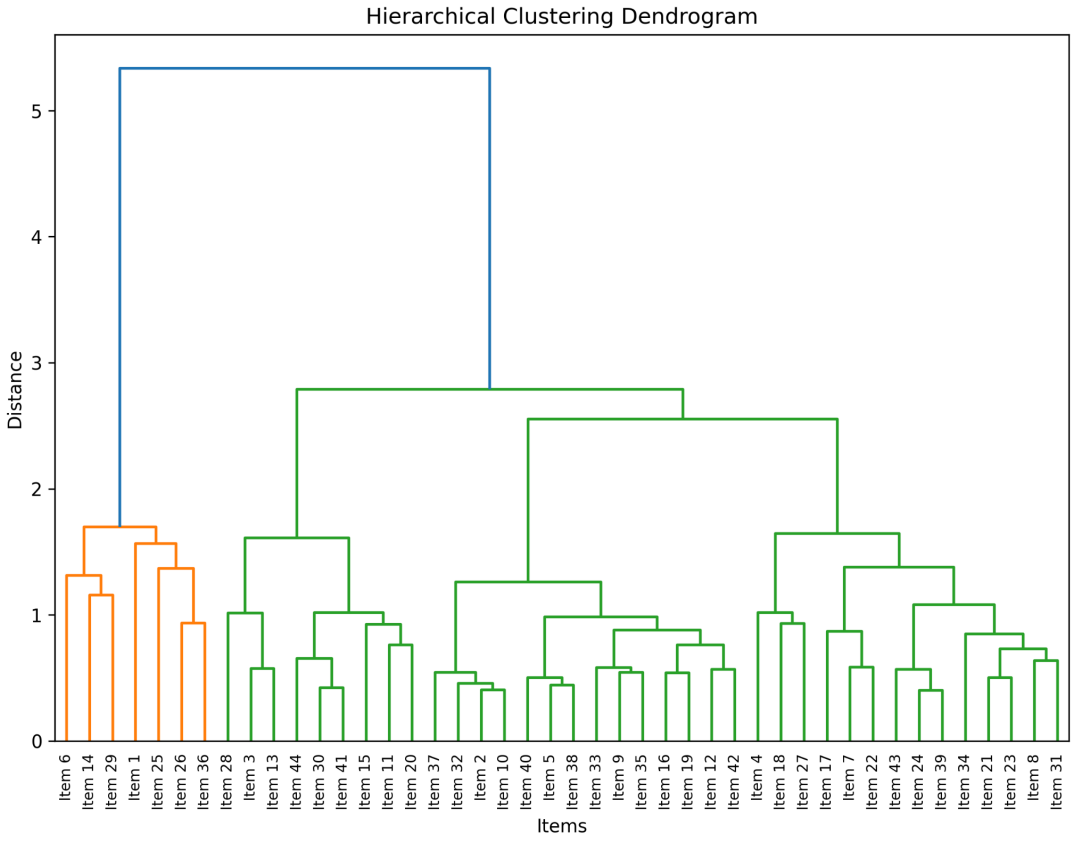


Figure 2: Hierarchical Clustering with GloVe

Question	True Trait	Spectral Trait	Hierarchical Trait
Is talkative	0	0	3
Tends to find fault with others	1	1	0
Does a thorough job	2	2	4
Is depressed, blue	3	3	2
Is original, comes up with new ideas	4	1	0
Is reserved	0	1	1
Is helpful and unselfish with others	1	1	2
Can be somewhat careless	2	3	2
Is relaxed, handles stress well	3	2	0
Is curious about many different things	4	1	0
Is full of energy	0	1	4
Starts quarrels with others	1	1	0
Is a reliable worker	2	2	4
Can be tense	3	3	1
Is ingenious, a deep thinker	4	4	4
Generates a lot of enthusiasm	0	1	0
Has a forgiving nature	1	1	2
Tends to be disorganized	2	3	2
Worries a lot	3	1	0
Has an active imagination	4	4	4
Tends to be quiet	0	3	2
Is generally trusting	1	1	2
Tends to be lazy	2	3	2
Is emotionally stable, not easily upset	3	2	2
Is inventive	4	4	3
Has an assertive personality	0	4	3
Can be cold and aloof	1	3	2
Perseveres until the task is finished	2	2	4
Can be moody	3	3	1
Values artistic, aesthetic experiences	4	4	4
Is sometimes shy, inhibited	0	3	2
Is considerate and kind to almost everyone	1	1	0
Does things efficiently	2	2	0
Remains calm in tense situations	3	3	2
Prefers work that is routine	4	2	0
Is outgoing, sociable	0	0	3
Is sometimes rude to others	1	1	0
Makes plans and follows through with them	2	1	0
Gets nervous easily	3	2	2
Likes to reflect, play with ideas	4	1	0
Has few artistic interests	4	4	4
Likes to cooperate with others	1	1	0
Is easily distracted	2	2	2
Is sophisticated in art, music, or literature	4	4	4

Figure 3: Overall result comparison with GloVe

ChatGPT

The analysis using ChatGPT (GPT-4o model) provided valuable insights into the semantic relationships between the 44 items in the BFI. A similarity matrix was constructed by averaging 30 semantic similarity scores for each pair of items, as rated by ChatGPT. This similarity matrix served as the foundation for clustering analysis and visualization.

The similarity matrix was visualized as a heatmap (Figure 3). Each cell represents the average similarity score between two items in the BFI. The diagonal of the heatmap shows perfect similarity scores (1.0), as each item is identical to itself. Clusters of higher similarity scores, which are represented by warmer colors (reds and oranges), show patterns where items likely belong to the same Big Five personality trait category.

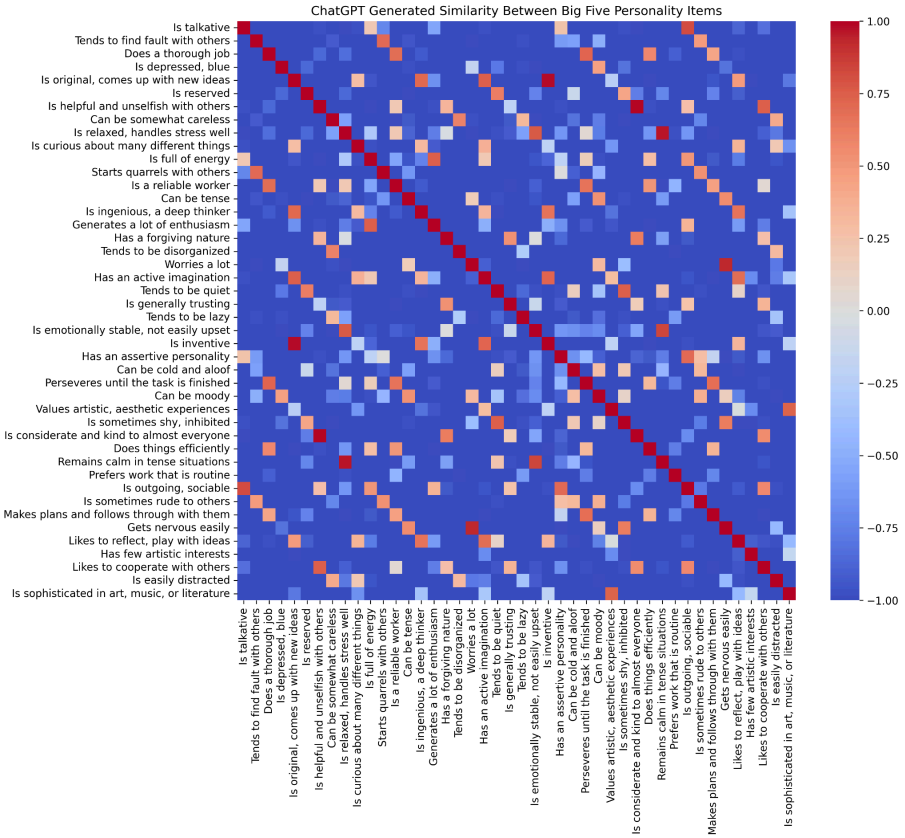


Figure 3: Similarity Matrix Generated by GPT-4o

Using the similarity matrix generated by ChatGPT, clustering analyses were then performed to classify the 44 items into five groups. Spectral clustering achieved a clustering accuracy score of 0.70. This indicates a moderately good alignment between the clusters identified by the model and the ground truth. The spectral clustering process revealed several clear groupings within the similarity matrix. Items with high semantic similarity, such as “is inventive” and “is original, comes up with new ideas” (associated with Openness), were frequently grouped together. Similarly, items tied to Agreeableness, like “is helpful and unselfish with others” and “has a forgiving nature”, are also strongly related. However, there’s also misclassifications in the results, especially for items with overlapping semantic meanings across

different traits. For example, “Is depressed, blue” could be related to both reversed extraversion and neuroticism.

Question	Cluster	Category	Trait	Ground Truth Category	Ground Truth Trait
Is talkative	0	0	Extraversion	0	Extraversion
Tends to find fault with others	0	0	Extraversion	1	Agreeableness
Does a thorough job	3	2	Conscientiousness	2	Conscientiousness
Is depressed, blue	0	0	Extraversion	3	Neuroticism
Is original, comes up with new ideas	1	4	Openness	4	Openness
Is reserved	0	0	Extraversion	0	Extraversion
Is helpful and unselfish with others	2	1	Agreeableness	1	Agreeableness
Can be somewhat careless	4	2	Conscientiousness	2	Conscientiousness
Is relaxed, handles stress well	2	1	Agreeableness	3	Neuroticism
Is curious about many different things	1	4	Openness	4	Openness
Is full of energy	0	0	Extraversion	0	Extraversion
Starts quarrels with others	0	0	Extraversion	1	Agreeableness
Is a reliable worker	3	2	Conscientiousness	2	Conscientiousness
Can be tense	0	0	Extraversion	3	Neuroticism
Is ingenious, a deep thinker	1	4	Openness	4	Openness
Generates a lot of enthusiasm	0	0	Extraversion	0	Extraversion
Has a forgiving nature	2	1	Agreeableness	1	Agreeableness
Tends to be disorganized	4	2	Conscientiousness	2	Conscientiousness
Worries a lot	0	0	Extraversion	3	Neuroticism
Has an active imagination	1	4	Openness	4	Openness
Tends to be quiet	0	0	Extraversion	0	Extraversion
Is generally trusting	2	1	Agreeableness	1	Agreeableness
Tends to be lazy	4	2	Conscientiousness	2	Conscientiousness
Is emotionally stable, not easily upset	2	1	Agreeableness	3	Neuroticism
Is inventive	1	4	Openness	4	Openness
Has an assertive personality	0	0	Extraversion	0	Extraversion
Can be cold and aloof	0	0	Extraversion	1	Agreeableness
Perseveres until the task is finished	3	2	Conscientiousness	2	Conscientiousness
Can be moody	0	0	Extraversion	3	Neuroticism
Values artistic, aesthetic experiences	1	4	Openness	4	Openness
Is sometimes shy, inhibited	0	0	Extraversion	0	Extraversion
Is considerate and kind to almost everyone	2	1	Agreeableness	1	Agreeableness
Does things efficiently	3	2	Conscientiousness	2	Conscientiousness
Remains calm in tense situations	2	1	Agreeableness	3	Neuroticism
Prefers work that is routine	4	2	Conscientiousness	4	Openness
Is outgoing, sociable	0	0	Extraversion	0	Extraversion
Is sometimes rude to others	0	0	Extraversion	1	Agreeableness
Makes plans and follows through with them	3	2	Conscientiousness	2	Conscientiousness
Gets nervous easily	0	0	Extraversion	3	Neuroticism
Likes to reflect, play with ideas	1	4	Openness	4	Openness
Has few artistic interests	1	4	Openness	4	Openness
Likes to cooperate with others	2	1	Agreeableness	1	Agreeableness
Is easily distracted	4	2	Conscientiousness	2	Conscientiousness
Is sophisticated in art, music, or literature	1	4	Openness	4	Openness

Table 1: Complete Clustering Results of Spectral Clustering Based on the Similarity Matrix Generated by GPT-4o

Hierarchical clustering was also applied to the similarity matrix to uncover relationships among items in a nested, tree-like structure. The resulting dendrogram as shown in Figure 4 illustrated the step-by-stop merging of items and clusters, which provide a detailed visualization of how semantic relationships unfold across different levels of similarity.

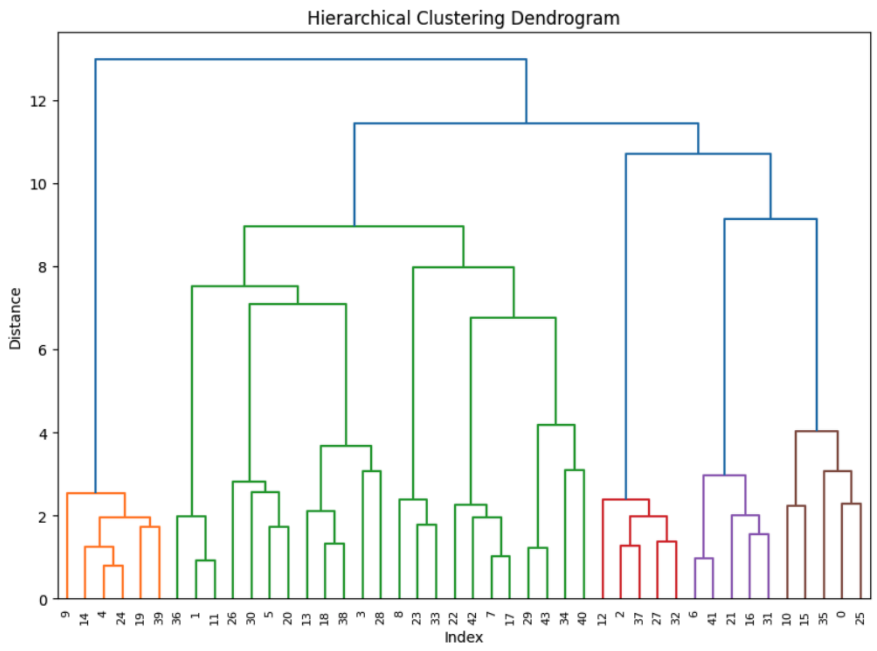


Figure 4: Hierarchical Clustering of the Similarity Matrix Generated by GPT-4o

The hierarchical clustering accuracy score was 0.66, which was slightly lower compared to spectral clustering. This is partly due to the rigid tree structure imposed by hierarchical clustering, which sometimes forced semantically ambiguous items into less fitting groups. Despite this limitation, the figure showed clear separations among traits with distinct and cohesive semantic meanings, such as Openness, where items like “is talkative” (1) and “is full of energy” (11) were grouped together. At the same time, traits with nuanced or context-dependent interpretations, like items belonging to Agreeableness and Neuroticism, were likely to intertwine with other categories, for example “likes to cooperate with others”.

Overall , the clustering accuracy scores of 0.70 (spectral clustering) and 0.66 (hierarchical clustering) indicates moderate success of this model in replicating the intended structure of the BFI based on semantic relationships alone. Both methods showed that ChatGPT’s similarity matrix could identify meaningful relationships between the items. However, it was still challenging to match the clusters perfectly with the actual personality

categories. Items with overlapping or flexible meanings were harder to classify, showing how complex it can be to group psychometric data. Despite these challenges, the results demonstrate that generative AI has potential as a tool for analyzing psychometric tests, offering a scalable and effective way to organize and evaluate test items.

GPT-Generate New Test Items

ChatGPT was also used to generate five new test items to address the third research question - whether genAI can produce new test items that align with the expected personality categories. These items were designed to correspond to the Big Five traits. The newly generated items are as follows:

- Enjoys being the center of attention in group settings (Extraversion)
- Tends to initiate conversations with strangers (Extraversion)
- Easily sympathizes with others' struggles (Agreeableness)
- Tries to avoid arguments, even when strongly disagreeing (Agreeableness)
- Double-checks work to ensure there are no mistakes (Conscientiousness)

These new items were added to the similarity matrix, expanding it to include 49 items in total. The updated matrix was then used to re-run both spectral clustering and hierarchical clustering to evaluate where the generative AI-produced items aligned with the intended categories.

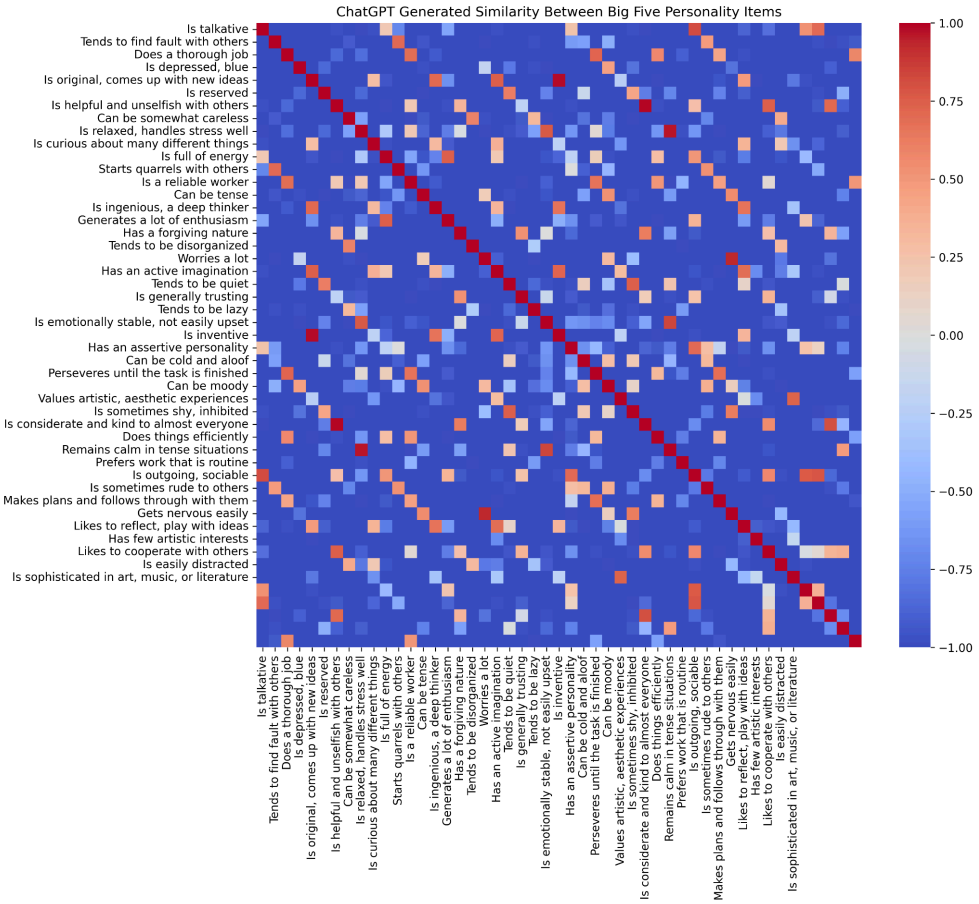


Figure 5: Updated Similarity Matrix Generated by GPT-4o

The result shows an improvement in clustering accuracy for both methods. Spectral clustering achieved an accuracy score of 0.76. This improvement indicates that the newly generated items contributed to a stronger overall grouping of the test items. Additionally, 4 out of the 5 new items were correctly classified into their expected personality trait categories. It misclassified the fourth item to the category Neuroticism. But in general, this improvement in the accuracy suggests that ChatGPT can effectively capture the semantic characteristics of the Big Five dimensions in these items.

Question	Cluster	Category	Trait	Ground Truth Category	Ground Truth Trait
Enjoys being the center of attention in group settings	1	0	Extraversion	0	Extraversion
Tends to initiate conversations with strangers	1	0	Extraversion	0	Extraversion
Easily sympathizes with others' struggles	4	1	Agreeableness	1	Agreeableness
Tries to avoid arguments, even when strongly disagreeing	0	3	Neuroticism	1	Agreeableness
Double-checks work to ensure there are no mistakes	3	2	Conscientiousness	2	Conscientiousness

Table 2: Complete Clustering Results of the 5 AI-Generated Test Items

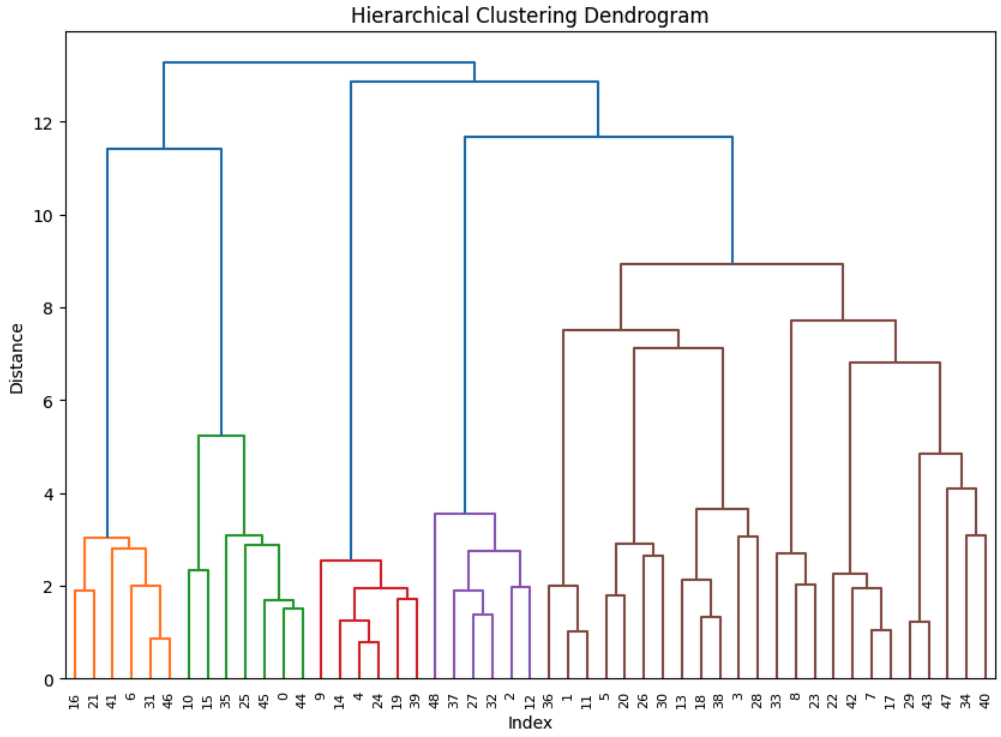


Figure 6: Updated Hierarchical Clustering of the Similarity Matrix Generated by GPT-4o

Discussion

RQ1: Convergence

The results suggest potentials in using language models to measure personality inventory validities. GloVe demonstrated approximately 50% accuracy in classifying the 5 categories, while ChatGPT achieved an accuracy as high as 70% in measuring convergence to the items' true categories. While GloVe was able to identify with fine accuracy in spectral clustering, it struggled to extract all five categories in hierarchical clustering. Unlike the 2 clusters produced

by GloVe's similarity matrix, the hierarchical clustering based on ChatGPT's similarity matrix showed 5 distinct clusters, which matched the original Big Five personality traits. From this aspect, ChatGPT outperforms GloVe as it can effectively capture the underlying structure of the BFI categories, also suggesting that it has a stronger capability of understanding and mirroring the theoretical foundation.

RQ2: Generative Capability

Our findings show that generative AI has the potential to not only analyze existing psychometric test data but also help expand them. By using ChatGPT, we created five new test items, added them to the similarity matrix, and evaluated the clustering performance with the updated data. The spectral clustering accuracy improved from 0.70 to 0.76, showing that the new items strengthened the overall clustering structure. Furthermore, 4 out of the 5 new items were correctly classified. This high accuracy confirms that ChatGPT was able to understand and capture the semantic essence of the Big Five traits in most cases. Hierarchical clustering also improved slightly, from 0.66 to 0.67, which also indicates that the new items provided additional information and clarity in the clustering process.

These results highlight both the strengths and challenges of using genAI for creating new psychometric test items. On the positive side, the generated items showed strong semantic coherence and contributed positively to the overall clustering performance. However, the misclassification of one item shows that AI-generated content does not always align perfectly with human-defined categories. This difference could be due to subtle ambiguities in language or the complexity of mapping semantic relationships to psychometric constructs.

Conclusion

This study utilized word embedding models (GloVe) and generative AI (ChatGPT) to measure the convergence of the Big Five Inventory (BFI) and use ChatGPT to generate new test items. Our findings highlight the strengths and potential of generative AI in psychometric research as it consistently outperformed GloVe in clustering accuracy, showing its ability to better capture nuanced and subtle relationships between different test items. These results also suggest that generative AI may be a helpful tool to provide deeper semantic understanding for analyzing and expanding psychometric tests.

Despite its strength, the subjective nature of GenAI's output can lead to inconsistencies of misclassification, as seen in one of the generated test items. Additionally, we're uncertain about the generalizability of these findings to other psychometric models as we only use one set of BFI in the study. The general performance of generative AI needs to be further compared with traditional methods like factor analysis to establish its reliability and validity more comprehensively.

Future research could extend these methods to other psychometric frameworks beyond the Big Five Inventory by exploring their applicability in various psychological constructs. Longitudinal studies could also help evaluate the stability and effectiveness of AI-generated items over time. In addition, ethical considerations, like mitigating biases in genAI outputs, must be addressed to ensure fair and responsible applications in psychometric testing.

In summary, this study shows that generative AI has the potential to change and even revolutionize psychometric research by offering scalable and effective solutions for analyzing and expanding personality assessments. However, careful evaluation and ongoing refinement are necessary to maximize its impact while addressing its limitations.

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